
Software Requirements Specification

for

Ride With Me Mobile Application

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document defines the requirement specifications for the **Ride With Me** Application version 1.0 that will enable users with vehicles to make some extra income and/or minimize cost in form of fuel while traveling on long journeys. This document covers the entire application.

1.2 Document Conventions

The font used in this document is Times New Roman, of size 12, line spacing of 1.5. The document is also justified.

Section headings are indicated by whole numbers (1, 2, 3...), and subsection headings are numbered based on the whole numbers of their sections for example; sub-section 1.1 belongs to section 1. Both sections and subsections are written with a bold-faced text for emphasis. *Italic* text is used to label diagrams and interfaces.

1.3 Intended Audience and Reading Suggestions

This document is to be read by the project managers, the development team, the documentation writers, marketing staff, users, testers, and the organization stakeholders associated with providing the necessary software.

The project managers and the development team need to become well versed with the requirements of the document, most importantly for the developers so they can effectively implement those requirements.

The marketing staff should understand the internal and external interface requirements of this document to become used to the various product features in order to effectively advertise the product. The testers need to be well-versed with the system features of the document to develop meaningful test cases and give useful feedback to the developers.

The stakeholders associated with providing the necessary software need to review the functional and non-functional requirements of this document to know what software will be required.

1.4 Product Scope

The Ride With Me Application will enable people with vehicles to make some extra income and/or minimize cost in form of fuel when traveling on long journeys. It will be a mobile-based application.

1.4.1 Goals

- Increasing income of car owners by earning extra money while traveling over long distances
- Reducing the costs of the journey on the car owners by sharing the costs.
- Increasing the number of safe trips taken by passengers.
- Making traveling convenient and stress-free for passengers, regardless of their location.
- To have no passenger waiting for longer than a couple of minutes for a car.
- To provide convenient and reasonably priced rides for anyone in need.

1.4.2 Objectives

Main Objective

- To optimize cost and allow owners of personal vehicles earn extra income while providing convenient, quick, safe and stress-free rides or trips.

Other Objectives

- To collect and analyze requirements for the Ride With Me Application.
- To design a model for an integrated Ride with Me Application.
- To implement the Ride With Me Application based on the design models.
- To test the Ride With Me Application with users generally; personal car owners and transportation companies.

1.4.3 Benefits

- Owners of cars will earn extra income from their vehicles.
- Passengers can travel more comfortably than they would in public transport means.

- Convenience and flexibility for both the passenger and the vehicle owner.
- Competitive transport fares.

1.5 References.

[1] Peter Cohen, Robert Hahn, Jonathan Hall, Steven Levitt, Robert Metcalfe (2016). Using Big Data to Estimate Consumer Surplus: The Case of Uber. NBER Working Paper No. 22627. <https://www.nber.org/papers/w22627>.

[2] Lee, Dave (2019-03-29). "For Uber and Lyft, reality is arriving soon". BBC. Archived from the original on 2019-07-27. Retrieved 2019-10-12

[3] Ronayne, Kathleen (2019-08-29). "Rideshare, delivery apps pledge \$90M California ballot fight". AP NEWS. Retrieved 2019-10-12.

[4] FREED, BENJAMIN (June 30, 2015). "Why You Shouldn't Call Uber and Lyft "Ride-Sharing"". Washingtonian.

[5] Concept paper for Ride With Me application

2. Overall Description

This section gives background information about specific requirements of the Ride With Me application to be developed. This section describes the factors that affect the final product.

2.1 Product Perspective

The following diagram gives a high-level overview of the system.

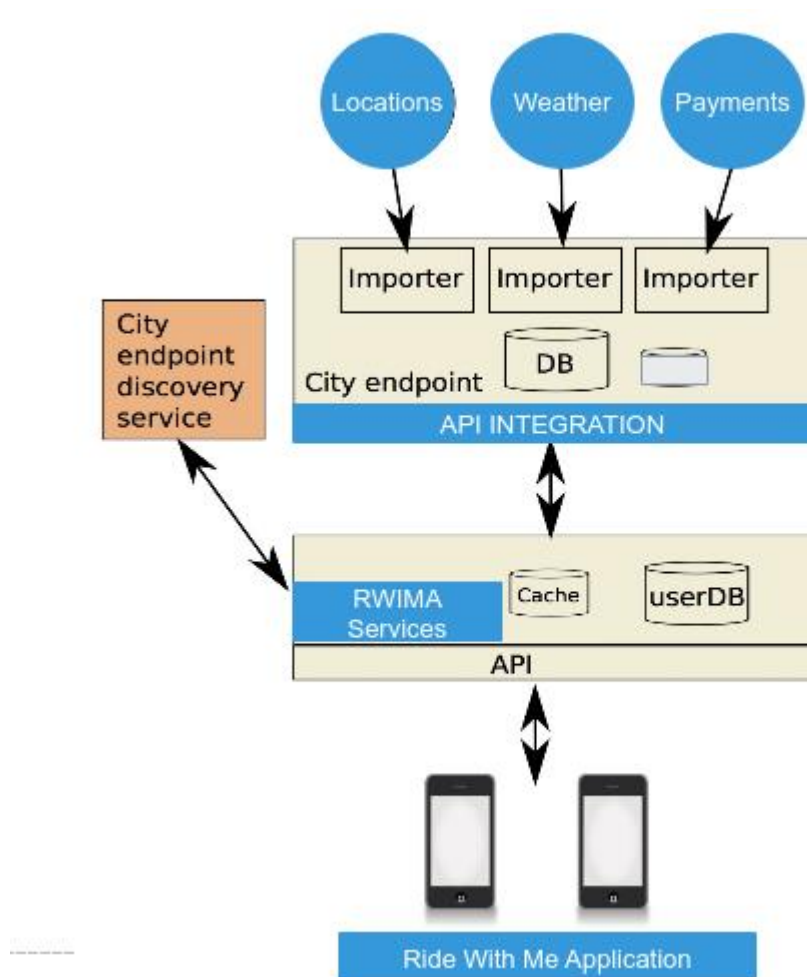


Figure 1: A block diagram showing the major components of the system.
Diagram showing the Major Components of the System.

Figure1: A Block

The Ride With Me app and the RWIMA service are within the scope of this document. The City endpoint and discovery service are also developed in the mobility work package, but should be regarded as external from this document's point of view.

The Ride With Me Application is a software which will allow various users to book and schedule rides.

The Ride With Me application is a special order software system. It will be used in the stated configuration.

2.1.1 Site Adaptation Requirements

Ride With Me Application is able to perform in every platform for example any browser that can be run on any operating system so it does not need any adaptation to a particular platform.

2.2 Product Functions

The major functions of Ride with Me Application are summarized below:

- **Create account:** New users need to create an account with the application in order to access its services.
- **Log in:** After creating an account, the user can sign in with their username and password.
- **Add Route:** Drivers may add a route by specifying, day and time of departure and number of empty seats. The user can select the route by two different way. The first way is entering start and end locations. Thus, the route is drawn on the map. The other way is selecting start and end locations on the map.
- **Select journey:** A passenger can select a journey they are taking depending on the route specified and advertised by a specific driver they choose.
- **Cancel journey:** A passenger can cancel a journey due to unavoidable reasons.
- **Delete route:** A driver may delete a route. After deleting route, passengers in that had selected that driver are informed by the system.
- **Make/ Receive calls:** The users can communicate to each other by making and receiving calls.
- **Rate Driver:** At the end of the journey, the respective passenger can rate the driver on the application. Thus, other interested passengers can see the driver rates and can decide to select that driver.
- **Register Driver:** The system's administrator enters the details of the driver into the system.

- **Accept journey request:** The driver accepts passenger's request by clicking "Accept" on the provided interface.

Use case diagram for ride with me mobile application

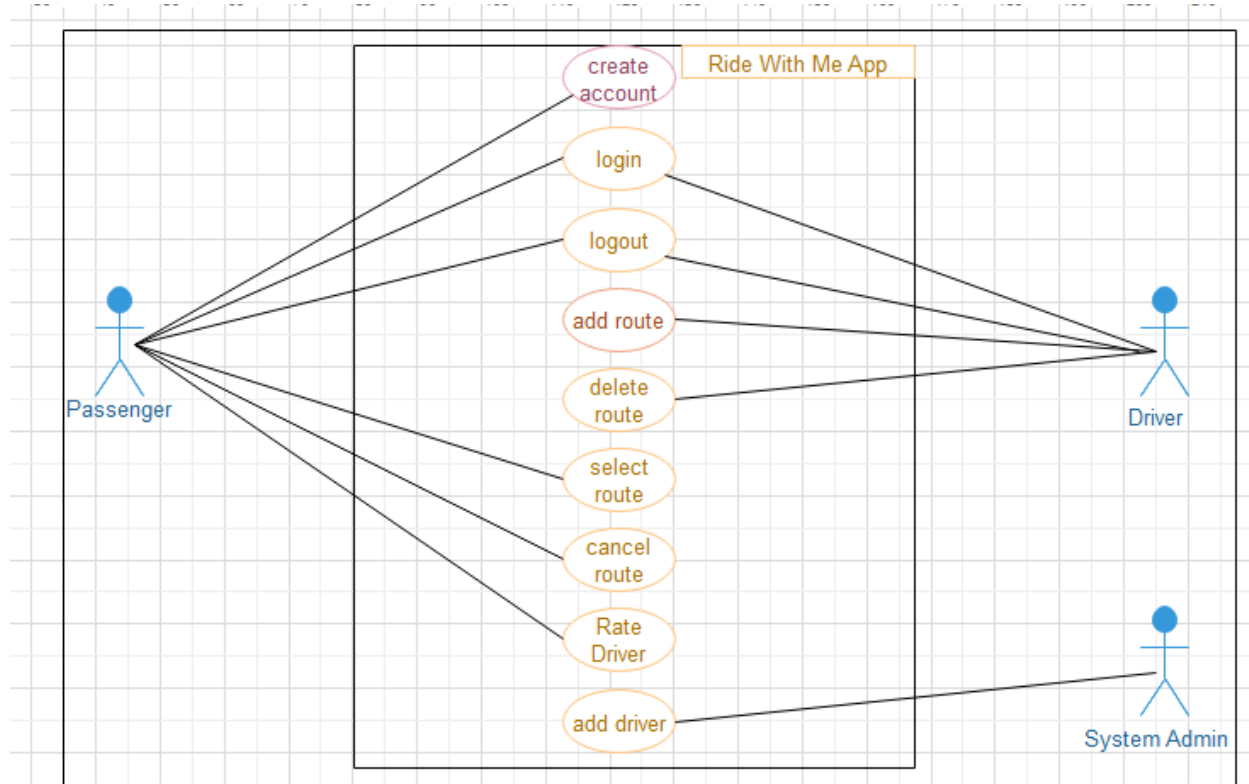


Figure 2: Use case diagram for Ride With Me Mobile Application.

2.3 User Classes and Characteristics

Below are the users of the Ride with Me Application and they are separated in 2 user classes;

2.3.1 Novice users.

These are the first time users of the system. Every user of the system (passenger or driver) is a novice user before and when they create an account with the system.

Characteristics

- They don't know their way around the system.
- They are first time users of the system.
- Require a lot of help and guidance to use the system.

2.3.2 Expert users.

When logging into personal accounts and performing each their distinct tasks, both passengers and drivers are expert users. For our system, these include; the system's administrator and users that have used the system before.

Characteristics

- They know their way around the system.
- They are not first time users of the system.
- Require less or no help to use the system.

2.4 Operating Environment

System can run on any smartphone with 1GB RAM and above and 8GB internal memory.

For computers, the system can run on any operating system;

The system can also be accessed through these browsers, brave browser, Firefox browser, Google chrome, opera's browser, sumsung internet browser.

2.5 Design and Implementation Constraints

There are a number of constraints which the system will abide by during development. And these include;

- System is to be developed for real time updates to make it more accurate and secure.
- System is to be developed in Java language.
- The design procedure involves identifying and assembling all the required hardware and ensuring safe interfacing between all the components.
- The developer will be required to use a machine with 8GB RAM for faster implementation of the system's functions.

- The system's response time should be 3 seconds or less.
- Another constraint is that the system requires remote server which enables the system functionality and data storage. Because of this situation, when the server crashes the system will not be able to resume its operations until the server become available to respond system requests. In addition to these, since the user information is stored in a database and this database can be hacked and user information will no longer be private to the user.
- After the user downloads the application onto their device, they are then responsible for maintaining it which includes; updating it if there are software updates available.

2.6 User Documentation

Here are some of the official documentations that will be delivered along with the system where the user can get more information about the system.

- Software design document which shows how the software system is structured to satisfy the user requirements.
- User manual which is a technical communication document intended to give assistance to people using the system.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

Below are some of the assumptions of the Ride with Me system;

- The user has a working smartphone or a computer.
- The user has an internet connection on the device they are using to access the system.
- Users will have access to a properly functioning vehicle with sufficient fuel.
- User interface and some functionalities can change during the development process of the project.
- New functionalities can be added which is able to change the dependent system requirements.

2.7.2 Dependencies

Below are some of the dependencies of our system;

- The system will be developed depending on the vision and scope explained in the system's concept paper.
- The system will depend on the fact that the developers to implement it are knowledgeable about the programming languages specified in the software interfaces.

3. External Interface Requirements

3.1 User Interfaces

The main interface of the Ride With Me App

The screenshot displays the main interface of the Ride With Me Application. On the left is a dark sidebar with a white 'RWIMA' logo and the text 'Ride With Me Application'. Below this is a user profile section with a green circular icon and the text 'User Full Name[Fulltitle]'. The sidebar contains a list of menu items: 'Manage Drivers' (highlighted in green), 'Manage Routes', 'Manage Passengers', 'Manage Requests', 'Manage Finance', 'System Settings', and 'Plans/Comments/suggestions'. The main content area has a light gray background. At the top of the main content area is a navigation bar with buttons for 'Home', 'Contact', 'Notifications', and 'User Profile'. Below this is a 'Manage Drivers' section with a 'Home' button. The central part of the main content area is a form for adding a driver, with fields for 'First Name', 'Last Name', 'Driver License', 'National Id', 'Vehicle Transport No', 'Phone Number', 'Guarantor', 'Guarantor Contact', 'Next Of Kin', and 'Next Of Kin Contact'. A blue 'Add Driver' button is at the bottom of the form.

Figure 3: The main interface of the Ride With Me Application.

This software product is developed for drivers and passengers. Product will be deployed to web site and on Google play store (for Android users), Apple store (for iOS users), and all users of the system will access the system through the web interface and/or mobile interface which includes multiple pages according to the system functionality for example for sign-up functionality, there will be a page for the creation of a new user account.

After the sign up, user information will be transferred to the database. In the sign up process, there will be an email verification to verify user information. After that point, users can log into their accounts/access the system through the web interface.

To access the system, every user has a unique username and password. In addition, there will be a database that stores and manipulates all the data about the users. The above interfaces will only be the interfaces for all the user data which stored by database and the execution of provided functionalities.

After log in, user will be able to log out whenever he or she wants.

1. This screen is to be used by the users of application for signing up. It will capture clear identity of passenger.

Thanks For Downloading Ride with Me Application
Please Signup

Enter First Name

Enter Last Name

Enter Email

Enter Phone Number

Signup

Figure 4: The interface used to capture user details during sign up.

2. Driver Screen

It will be used passengers to select drivers heading to a specified location.

The driver Screen



Figure 5: The Driver's interface.

3.2 Hardware Interfaces

Because Ride With Me Application is web based, it is compatible with all the browsers and can be run on any operating system of either 64bit or 32bit.

Our system will run optimally on any computer with a minimum of 4 cores processor, a minimum of 8 GB RAM and at least 80 GB for system drive.

3.3 Software Interfaces

The system has Google Map API as a subsystem. Google Map subsystem has their own web based interface which is a map that consists of roads and locations in a desired area and the user can easily intersect with this system.

Wamp/Xampp software to access database management system is required software product for Ride With Me Application because all data about system for example user and route information must be stored in database for later use and system functionality. MySQL database management system is used for that purpose and it has a nice open source user interface which displays table and rows in well formatted form for developers to create and manage the whole database.

The system shall be developed using JavaScript, PHP and Bootstrap as a web-based system to illustrate the system attractively.

The system shall be developed using Android studio when translating the system into an Android App.

The system shall also use fire base online database for sending and receiving notifications

The system shall use the mobile money API for payment processing.

3.4 Communications Interfaces

The system shall send automatic verification link to the user's email address during the process of signing up in the system. Moreover, in communication between driver and passenger, users shall send and receive notifications through the Fire base real-time database.

4. System Features

This section describes the features of the Ride with Me Application;

4.1 Account management.

4.1.1 Create account.

4.1.1.1 Description and Priority

After downloading and installing the application on device, the system will allow the passenger to create an account by entering their first name, last name, phone number and email address. After submission of these details, a token is sent to the user's email and the user is prompted to verify their email address.

This is a high priority feature as the user cannot access the system without an account.

4.1.1.2 Stimulus/Response Sequences

Use case name	Create account
Use case ID	UC1
Priority	High
Actor(s)	1. Passenger 2. System
Description	This use case describes the event of a passenger creating a new account.
Preconditions	Passenger has downloaded and installed the system on his/her device.
Normal flow of events	1. The passenger opens the application on the device. 2. The passenger enters his/her first name, last name, phone number and email address. 3. The passenger clicks submit button to submit the details.

	4. The system sends an account activation and verification link to the passenger's email. 5. The passenger clicks on the link and h/she is prompted to set a password for h/she account. 6. The passenger types in a preferred password in the interface provided. 7. The passenger re-types in the password to verify. 8. The passenger clicks on a submit button to set the password. 9. The system notifies the passenger of the successful account creation and directs him/her to a login page.
Post conditions	Passenger has created an account with the system.
Alternative flows	Flow 1: 1. If one of the required fields (first name, last name, phone number, email address) in sign up page are not filled properly, a warning message will be shown by the system.

Table 1. Showing stimulus/response sequences for the use case create account.

4.1.1.3 Functional Requirements

REQ-1: The system shall enable a passenger to enter his/her details.

REQ-2: The system shall enable a passenger to activate his/her personal details.

REQ-3: The system shall enable a passenger to set a password for his/her account.

4.1.2 Login

4.1.2.1 Description and Priority

This feature will allow a user to login to the Ride With Me Application after creating an account with the company. This feature will be used by the passengers who will have installed the application on their devices.

4.1.2.2 Stimulus and Response

USE CASE ID	UC6
Actor	Passenger
Precondition	The user shall be able to sign to the system.
Post condition	The user will be able to use the system.
Precedence	Mandatory
Normal flow of events	The user opens the system and enters his or her login details. User presses the login button User enters his or her password
Alternative flows	Flow 1: If the user enters a wrong password he/she will be prompted to enter the correct information Flow 2: User will be directed to the appropriate screen

Table 2. Showing Stimulus/Response Sequences for passenger use case.

4.1.2.3 Functional Requirements

REQ-17: The system shall allow a user to login.

REQ-18: The system shall display available drivers on the route

REQ-19: The system shall check user credentials on login.

4.2 Route management.

4.2.1. Delete route

4.2.1.1 Description and Priority

After a driver has added a route and cannot take the journey due to unavoidable reasons, h/she can delete that route that /she was planning on taking. The system automatically removes the driver's username from the list of available drivers on that particular route. The system then notifies all passengers that had selected that driver.

4.2.1.2 Stimulus/Response Sequences

Use case name	Delete route
Use case ID	UC2
Priority	High
Actor(s)	1. Driver. 2. System.
Description	This use case describes the event of a driver deleting a route.
Preconditions	1. The driver has been added into the system by the system administrator. 2. The driver had added a route on his account.
Normal flow of events	Step1: The driver logs into his/her account. Step2: The driver goes to the interface of all accepted passenger requests. Step3: The driver clicks the "Delete route" button.

	Step4: The system removes the driver's name from the list of available drivers on that specific route. Step5: The system sends notifications to all the passengers whose requests had been accepted by that driver. Step6: Those passengers receive the notifications.
Post conditions	1. The driver's name has been removed from the list of available drivers on that specific route. 2. The system has sent notifications to all the passengers whose requests had been accepted by that driver.
Alternative flows	None.

Table 3. Showing Stimulus/Response Sequences for Delete route use case.

4.2.1.3 Functional Requirements

REQ-4: The system shall enable a driver to enter log into their account.

REQ-5: The system shall enable a driver to go to the interface for accepted passenger requests.

REQ-6: The system shall remove the driver's name from the list of available drivers on that specific route.

REQ-7: The system shall send notifications to all the passengers whose requests had been accepted by that driver.

4.2.2 Select journey.

4.2.2.1 Description and Priority

This feature is used by a passenger when he/she is selecting a journey that h/she may like to take. This involves them selecting the required route and then a driver they prefer depending on the driver's specifications. The system then sends a passenger request to the respective driver who decides whether to accept it or reject it. The system then sends an appropriate notification to the respective passenger. This feature is high priority.

4.2.2.2 Stimulus/Response Sequences.

Use case name	Select journey.
Use case ID	UC3
Priority	High
Actor(s)	1. Driver. 2. System.
Description	This use case describes the event of a passenger selecting a journey to make.
Preconditions	1. The passenger has an account in the system. 2. The driver has logged into his/her account.
Normal flow of events	Step1: The passenger selects a route he/she wants to take. Step2: The passenger looks at available drivers' specifications. Step3: The passenger selects a preferred driver. Step4: The system sends passenger request to the respective driver. Step5: The driver receives passenger request.

	5.1: Driver accepts request. Step6: The system sends an acceptance notification to the passenger.
Post conditions	1. The passenger has selected a preferred driver. 2. The passenger's request has been accepted by the respective driver.
Alternative flows	Flow1: Step5: 5.1: The driver rejects the passenger request. 5.2: The system sends the respective passenger a rejection notification. 5.3: The respective passenger receives the notification.

Table 4. Showing Stimulus/Response Sequences for select journey.

4.2.2.3 Functional Requirements

REQ-8: The system shall enable a passenger to select a specific route.

REQ-9: The system shall enable a passenger to look at the details of all the available drivers taking that route.

REQ-10: The system shall send a passenger request to a respective driver.

REQ-11: The system shall enable the driver to either accept or reject a passenger request.

REQ-12: The system shall send either an acceptance or rejection notification to the respective passenger.

4.2.3 Cancel journey.

4.2.3.1 Description and Priority

This feature describes a situation where a passenger cancels a journey due to unavoidable reasons. After a driver has accepted the passenger's request and the system has sent an acceptance notification

to that respective passenger, the passenger may decide to cancel the journey by clicking on the “Cancel journey” button in that interface. This feature is of moderate priority.

4.2.3.2 Stimulus/Response Sequences

Use case name	Cancel journey.
Use case ID	UC4
Priority	Moderate
Actor(s)	1. Passenger. 2. System.
Description	This use case describes the event of a passenger canceling a journey they wanted to make.
Preconditions	1. The passenger has already received an acceptance notification.
Normal flow of events	Step1: The passenger goes to the interface of the acceptance notification. Step2: The passenger clicks on the “Cancel journey” button. Step3: The system sends a notification to the respective driver. Step4: The driver receives notification.
Post conditions	1. The passenger has canceled the journey. 2. The respective driver has received a notification on journey cancellation.
Alternative flows	None.

Table 5. Showing Stimulus/Response Sequences for cancel journey use case.

4.2.3.3 Functional Requirements

REQ-13: The system shall enable a passenger to cancel a journey.

REQ-14: The system shall send a journey cancellation notification to respective driver.

4.2.4 Accept journey request.

4.2.4.1 Description and Priority

This feature describes a situation where the system has sent journey request to the respective driver and the driver decides to accept the request. The driver clicks the “Accept ” button in the interface of that request. The system then sends an acceptance notification to the respective passenger.

4.2.4.2 Stimulus/Response Sequences

Use case name	Accept journey request.
Use case ID	UC5
Priority	High
Actor(s)	1. Driver. 2. System.
Description	This use case describes the event of a driver accepting a journey request.
Preconditions	1. The driver has already received a journey request.
Normal flow of events	Step1: The driver goes to the interface of the journey request. Step2: The passenger clicks on the “Accept request” button. Step3: The system sends an acceptance notification to the respective passenger. Step4: The passenger receives notification.
Post conditions	1. The driver has accepted the journey request. 2. The respective passenger has received an acceptance notification.
Alternative flows	None.

Table 6. Showing Stimulus/Response Sequences for accept journey request use case.

4.2.4.3 Functional Requirements

REQ-15: The system shall enable a passenger to accept a journey.

REQ-16: The system shall send a journey request acceptance notification to respective passenger.

4.2.5 Add route

4.2.5.1 Description and Priority

This feature will be used by the owner of the car to choose a route where he/she is heading to. Then prospective passengers can search drivers accordingly to where they are heading to.

4.2.5.2 Stimulus and Response

USE CASE ID	UC7
Actors	Driver
Description	Driver shall be able to add a route from the map.
Pre-condition	The user shall be able to sign to the system
Post condition	User must be able to receive transportation requests from other users.
Precedence	Mandatory
Normal flow of events	1. Drive shall enter his or her profile page 2. Driver shall add a new transportation route 3. Add transportation page will be loaded. 4. Driver enters available seats, departure time, type of car, number plate. 5. Driver draws a route on the map panel

Alternative Flow	Flow 1: 1. The user clicks the radio button of “one time”. 2. User enters date information in terms of day, month and year. Flow 2: 1. The user clicks the radio button of “periodic”. 2. User selects days from the week day check boxes.
------------------	--

Table 7. Showing Stimulus/Response Sequences for UC7.

4.2.5.3 Functional Requirements

REQ-20: The system shall allow a user to create a route of travel.

REQ-21: The system shall allow a user to set up a pick up locations while on a route.

REQ-22: The system shall allow a user to select a route.

REQ-23: The system shall display drivers profile when a passenger selects a journey.

4.3 Driver feature

4.3.1 Rate Driver

4.3.1.1 Description and Priority

A passenger will be provided with field to provide feedback to the Ride With Me Application developers the code of conduct of the driver. Will also provide additional information on skills and experience of driver when on the road. This feature is of high priority since it will help us on improvement of our services.

4.3.1.2 Stimulus and Responses.

USE CASE ID	UC3
Actor	Passenger
Description	User shall be able to rate the driver through the system.
Preconditions	The transportation route shall be completed.
Post conditions	Driver rating will be updated
Precedence	Not Mandatory
Normal flow of event	1. After the transportation, the hitchhiker login to the system. 2. Popup window is opened.
Alternative Flow(s)	Flow 1: 1. Hitchhiker clicks the star icon to rate driver's related attitude. Flow 2: 1. Hitchhiker clicks close icon. 2. The popup window will be closed

Table 8. Showing Stimulus/Response Sequences for UC3

4.3.1.3 Functional Requirements

REQ-24: The system shall pop up a window to assign a rating to a driver.

REQ-25: The system shall submit a rating to the system administrator for evaluation

4.3.2 Register Driver

4.3.2.1 Description and Priority

The system administrator registers the driver who meet all the requirements. Drivers will submit his credentials like driving permit via our online portal, then approval will be done accordingly after validating his details by the system administrator.

4.3.2.2 Stimulus and Response

USE CASE ID	UC4
Actor	Driver
Description	Use case shall register a driver
Precondition	Driver has valid credentials like driving permit
Post condition	Driver given his login credentials
Precedence	Mandatory
Normal flow of events	1. Driver submits his or her details for approval. 2. System admin validates the details of the driver. 3. An approval or rejection notice is sent via email.
Alternative flow	Flow 1: If the driver enters a wrong password or username he/she will be prompted to enter the correct information Flow 2: Driver will be directed to the driver screen

*Table 9. Showing Stimulus/Response Sequences for UC8.***4.3.2.3 Functional Requirements**

REQ-26: The system shall allow a driver to register.

REQ-27: The system shall notify a user the status of his transaction.

REQ-28: The system shall notify a user via email whether a transaction is approved or rejected.

Appendix A: Glossary

Terms	Definition
API	Application Programming Interface
etc.	And many others
GB	Gigabyte
GSM	Global System for Mobile Communication
GUI	Graphical User Interface
ID	Identification
IEEE	Institute of Electrical and Electronics Engineers
LCD	Liquid Crystal Display
MYSQL	My Structured Query Language
PHP	Hypertext Preprocessor
PIC	Peripheral Interface Controller
POST Request	A method used to send information to the database
RAM	Random Access Memory
REQ	Requirements
SRS	Software Requirement Specification
UC	Use case
User	Someone who interacts with the system

Table 10. Showing the terms and their definitions.